

Exact Joint Survival and Mixed Moments for Twenty First-Passage Targets

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Abstract

We compute the complete joint first-passage coupling induced by a uniform ordering of the sixteen frozen labels. For every ordered pair of the twenty actual subgroup targets, all 17×17 joint survival counts and reduced probabilities are exact. Nested support enumeration supplies the counts, while two-dimensional tail summation gives mixed moments through bidegree six and covariance. Every marginal is recovered in 400 independent pair checks. A separate bitset/zeta reconstruction, SymPy arithmetic, clean replay, and thirteen hostile mutations audit the receipt. The scope is `NO_BAD_EULER_OR_ROOT_NUMBER`.

1 Joint survival surface

Let T_i and T_j be C88 first-passage times and define

$$J_{ij}(k, l) = \#\{\pi : T_i(\pi) > k, T_j(\pi) > l\}, \quad 0 \leq k, l \leq 16.$$

For $k \leq l$, choose a non-hit k -support A for i and a non-hit l -support $B \supseteq A$ for j . The pair has $k!(l-k)!(16-l)!$ completions to a full permutation. The reverse order is obtained by transposition.

Theorem 1. *The canonical C90 receipt contains every $J_{ij}(k, l)$ for all 400 ordered pairs. With $\Delta_a(k) = (k+1)^a - k^a$ and $a, b \leq 6$,*

$$\mathbb{E}[T_i^a T_j^b] = \frac{1}{16!} \sum_{k, l=0}^{15} \Delta_a(k) \Delta_b(l) J_{ij}(k, l).$$

Orders $(a, 0)$ and $(0, b)$ recover the exact C89 marginal raw moments, and the covariance is $\mathbb{E}[T_i T_j] - \mathbb{E}[T_i] \mathbb{E}[T_j]$.

Proof. The first statement follows by partitioning permutations according to their two nested prefix supports. The displayed formula is the finite two-variable tail-sum identity; the marginal and covariance formulas are immediate specializations. $\square$

2 Exact inventory and audit

There are $400 \cdot 289 = 115600$ joint survival cells, $400 \cdot 49$ mixed moment cells, and 400 covariance values. The checker reconstructs all cells from C88 hit bitsets using an independent superset zeta transform. Pair transpose symmetry, diagonal single-target survival recovery, monotone tail surfaces, and both marginal axes are checked exactly. SymPy verifies every mixed moment and covariance. Clean replay preserves

`c457a267b2621c71f7f5ad810ce9dec41aacfe25de3e843fab1398be75571978.`

No arithmetic/local-data, Euler-factor, root-number, automorphy, full Burnside/table-of-marks, or Hilbert–Polya operator claim is made.